

Generated Question Paper

Q1. State two advantages of using a pyramid of energy rather than a pyramid of biomass to represent a food chain. [2 marks]

Q2. [3cm] (d) Shows a photomicrograph of a sperm cell. [3cm] (e) Draw and label one human egg cell, including at least one feature not found in a sperm cell. [3 marks]

■ Diagram not found: images/sperm_cell_micrograph.png]

Q3. [3cm] (c) Explain why plants store starch in potato tubers. [2 marks]

Q4. [1.5cm] (c) Explain the reasons for the different temperatures used in step 3 and step 5. [2 marks]

Q5. [1.5cm] (ii) State the name of the process occurring at step 3 that causes gas bubbles to form in the dough. [1 marks]

Q6. Enzymes are catalysts. (a) (i) Describe what is meant by the term catalyst. [2 marks]

Q7. (f) Describe what happens to a fertilised egg cell before implantation in the uterus. [3 marks]

Q8. [1.5cm] (b) Complete Fig. 5.1 by stating two other organs that belong to two organ systems. [1.5cm] (c) Fig. 5.2 shows photomicrograph of part of mammalian testis. Explain why meiosis is necessary in the testes. [3 marks]

■ Diagram not found: images/organs_two_systems.png]

■ Diagram not found: images/mammalian_testis_micrograph.png]

Q9. [5cm] (c) State the names of molecules converted into ammonium ions in the nitrogen cycle. [1 marks]

Q10. [4cm] (d) Scientists had higher breathing rates in the 550 ppm CO₂ plot. Suggest why. [2 marks]

Q11. (a) On Fig.2, label with an R the position of the right ventricle. [1 marks]

■ Diagram not found: images/Fig2_1_HeartCrossSection.png]

Q12. [1.5cm] (e) Explain the effects of nitrate ion deficiency on plant growth. [3 marks]

Q13. (ii) Fishing can be managed to promote the conservation of fish stocks. Explain one way that fishing can be managed to conserve fish stocks. [2 marks]

Q14. (d) State the name of the process converting ammonium ions into nitrate ions. [1 marks]

Q15. (a) State the name of organism A in the figure above. [1 marks]

[■ Diagram not found: images/bread_making_steps.png]

Q16. A: Blood vessel C. Blood vessel C transports blood from the to the . [2 marks]

Q17. (ii) State the names of the structures labelled A and B on Fig.2. [2 marks]

Q18. (b) State the organs that make up the central nervous system. [1 marks]

Q19. [1.5cm] (b) Lugworms live in sand on beaches and feed on diatoms, which require ammonium ions. Researchers tested effects of microplastics on lugworms; see Table 3.1. Explain and explain why researchers concluded that non-biodegradable microplastics are most harmful to lugworms using Table 3.1. [4 marks]

Q20. (ii) Enzymes are proteins. State the names of the chemical elements found in all proteins. [2 marks]

Q21. The effect of pH on the percentage activity of lipase is shown on Fig.1. [6 marks]

[■ Diagram not found: images/Fig1_1_LipaseActivity.png]

Q22. Table 5.1: Plants growing in Europe begincenter begintabularcc toprule Plant Number & Leaf Area (cm\$2\$) midrule 1 & 310 2 & 335 3 & 390 4 & 348 5 & 365 midrule Mean & 350 bottomrule endtabular endcenter (i) Using the data in Table 5.2, calculate the mean leaf area for plants growing in North America. Give your answer as a whole number and include the unit. [2 marks]

Q23. (d) State the names of two products of biotechnology, other than bread, that make use of microorganisms. [2 marks]

[illegible]

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..... [1 marks]

Q25. (e) Shows a synapse and parts of two neurons. (i) Draw an arrow to show the direction of the signal across the synapse. [1 marks]

[■ Diagram not found: images/synapse_diagram.png]

Q26. (d) Suggest why reflexes occur in unconscious people. [1 marks]

Q27. [2cm] (b) Fig. 6.2 shows a reproduction method used by some potato farmers. Explain the advantages of this type of reproduction in crop production. [3 marks]

[■ Diagram not found: images/potato_tubers_method.png]

Q28. [2cm] Very small pieces of plastic called microplastics are found in many products such as soaps and toothpaste. (a) It is estimated that microplastics make up 5 (ii) State one recommendation, other than regular brushing, for proper care of teeth. [1 marks]

Q29. (a) State the hormone released from the testes. [1 marks]

Q30. (a) The oceans contain fish stocks that can be managed as a sustainable resource to provide food for humans. (i) State what is meant by the term sustainable resource. [2 marks]